SMART CITY BIKE SHARING

1. Project Overview and Objective

Public bike-sharing systems generate continuous data from hundreds of stations across different cities. The task is to analyze this real-time bike station dataset to understand station performance, usage efficiency, and operational patterns across cities.

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2. Data Sources

- **Source:** Dataset link
- **Timeline: 2022 - 2025**
- **Domain**: Smart City Development
- **Description:**
    - The Dataset encapsulates vital statistics on Smart city Bike sharing data of a strategic assessment of a European micro-mobility network consisting of nearly 3000 active stations. The report evaluates the system through six core pillars: infrastructure distribution architecture, targeted incentive programs, hardware asset design, operational health, geographic flagship performance, and terminal journey risks.

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3. Problem Statement

- To find solution eliminate operational blind spots.
- To analyse how we can prevent return-journey bottlenecks.
- To evaluate and optimize hardware cost inefficiencies.
- To analyse how we can mitigate terrain-driven fleet imbalances.
- To analyse how to minimize concentrated flagship market risk.
- To analyse how do we transition to targeted maintenance.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Name	String (Text)	Name of the bike station in Europe
Address	String (Text)	Address of the Bike stations
Position	Decimal Number	Geographical Latitude, Longitude values for precise location
Banking	String (Text)	Describe the availability of mobile application adoption of Banking
Bonus	String (Text)	Describe the availability of Bonus Insentive
Status	String (Text)	Describe the status of the Bike station whether it is open or close.
Contract Name	String (Text)	Name of the Contract Team who owns the Bike station
Bike Stands	Whole Number	Total Number of Bike stands

Available Bike Stands	Whole Number	Number of available Bike stands
Available Bikes	Whole Number	Number of Available Bikes
Last Update	Date/Time/Timezone	Date and Time details with time zone detail

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5. Tools & Technologies

- **Power Query:** Data cleaning, transformations..
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Standardized formats, and created calculated fields and columns.
- **Filtering & Sorting:** Organized data to focus on relevant records.

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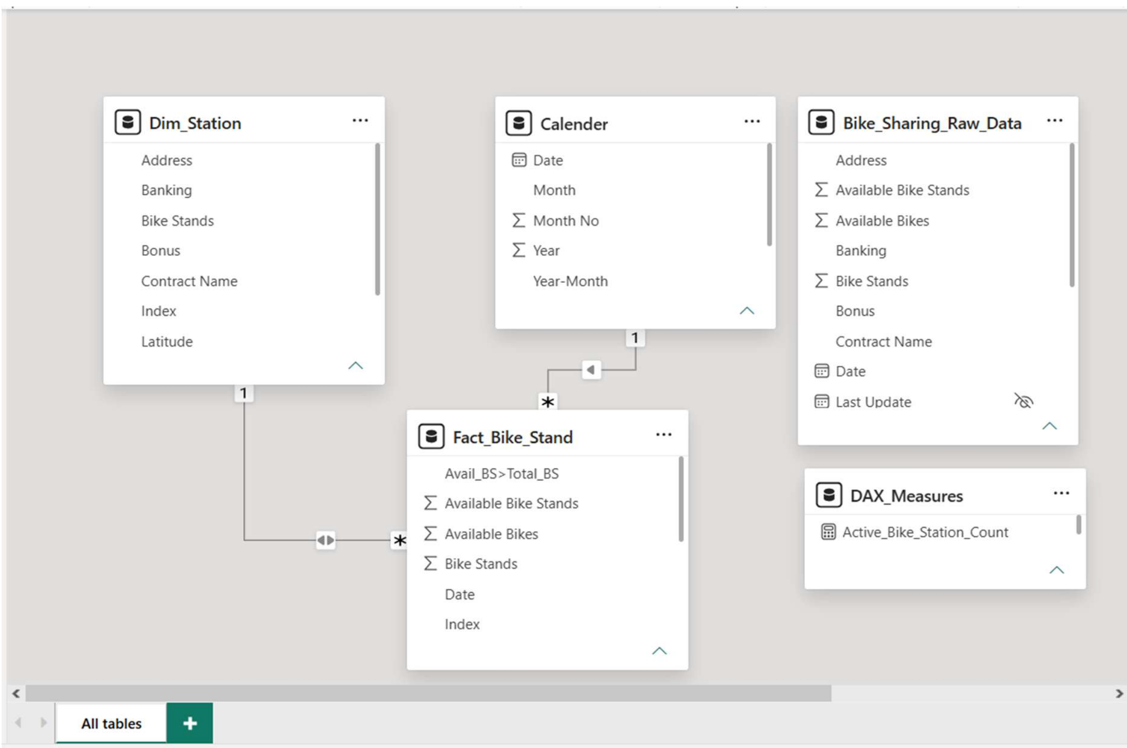
7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables Fact_Bike_Stand, Dim_Station and Date table defined cardinality.



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics are,
  - Active_Bike_Station_Count =

```
CALCULATE(
COUNTROWS(Fact_Bike_Stand),
Fact_Bike_Stand[Available Bike Stands] <> 0
)
```

- Avg Capacity - Banking App/Member only =

    CALCULATE(

    AVERAGE(Dim_Station[Bike Stands]),

    Dim_Station[Banking] = "FALSE"

)
- Avg Capacity - Banking Enabled =

    CALCULATE(

    AVERAGE(Dim_Station[Bike Stands]),

    Dim_Station[Banking] = "TRUE"

)
- Communication_Status =

    IF([Minutes_Since_Last_Update] > 30, "🚨 Stale Data (Offline)", "☑ Live Data (Online)")
- Inactive_Bike_Station_Count =

    CALCULATE(

    COUNTROWS(Fact_Bike_Stand),

    Fact_Bike_Stand[Available Bike Stands] = 0

)
- Last_Reported_Timestamp = MAX(Fact_Bike_Stand[Last Update])

- Percentage of Bonus Stations =

    DIVIDE(

    CALCULATE(

    COUNTROWS(Dim_Station),

    Dim_Station[Bonus] = "True"

    ),

    COUNTROWS(Dim_Station),

    0

    Last_Reported_Timestamp = MAX(Fact_Bike_Stand[Last Update])

)
- Percentage of Operational Stations =

    DIVIDE(

    CALCULATE(

    COUNTROWS(Fact_Bike_Stand),

    Fact_Bike_Stand[Status] = "OPEN"

    ),

    COUNTROWS(Fact_Bike_Stand),

    0

)
- Total_Network_Stands = SUM(Dim_Station[Bike Stands])

- Data model done by separate the raw data set into fact and dimension tables and Date table is Created.
- One Measurable table is create to align and identify all measures easily.
- One Calculated text column was created in Fact_Bike_Stand table to clean Name column
- One unnecessary column named Number had removed.
- Custom columns are added for separate Date and Time.
- Location column split into latitude and Longitude columns.
- Index column is created in fact and dimension tables.

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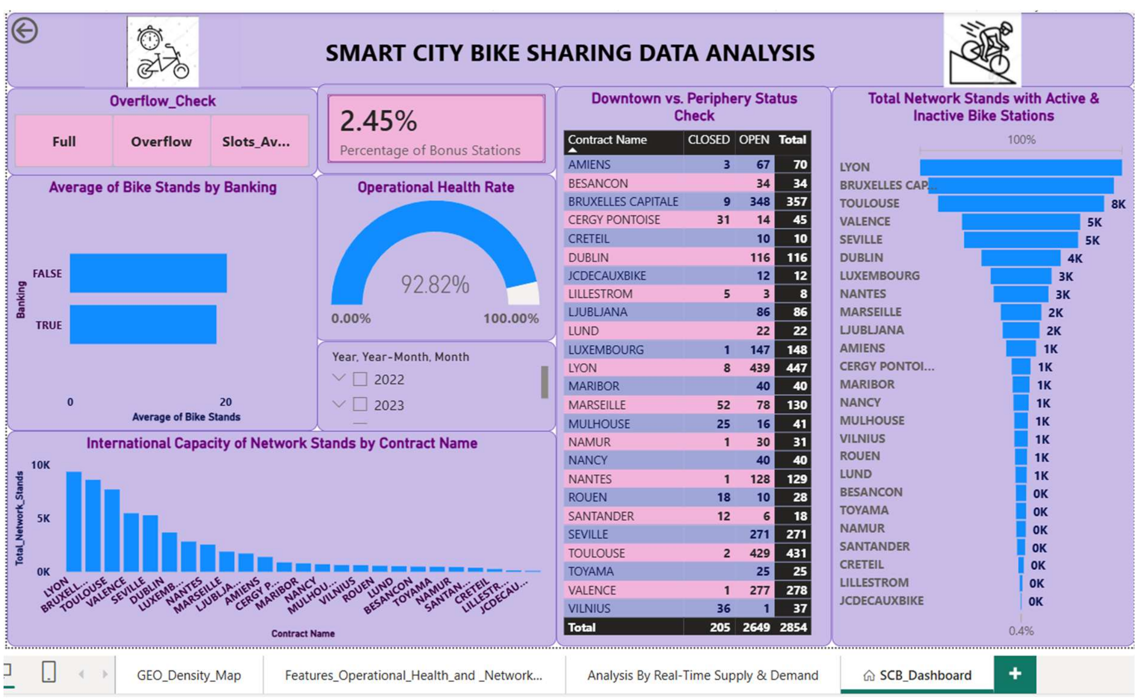
8. Analysis and Visualizations (Power BI)

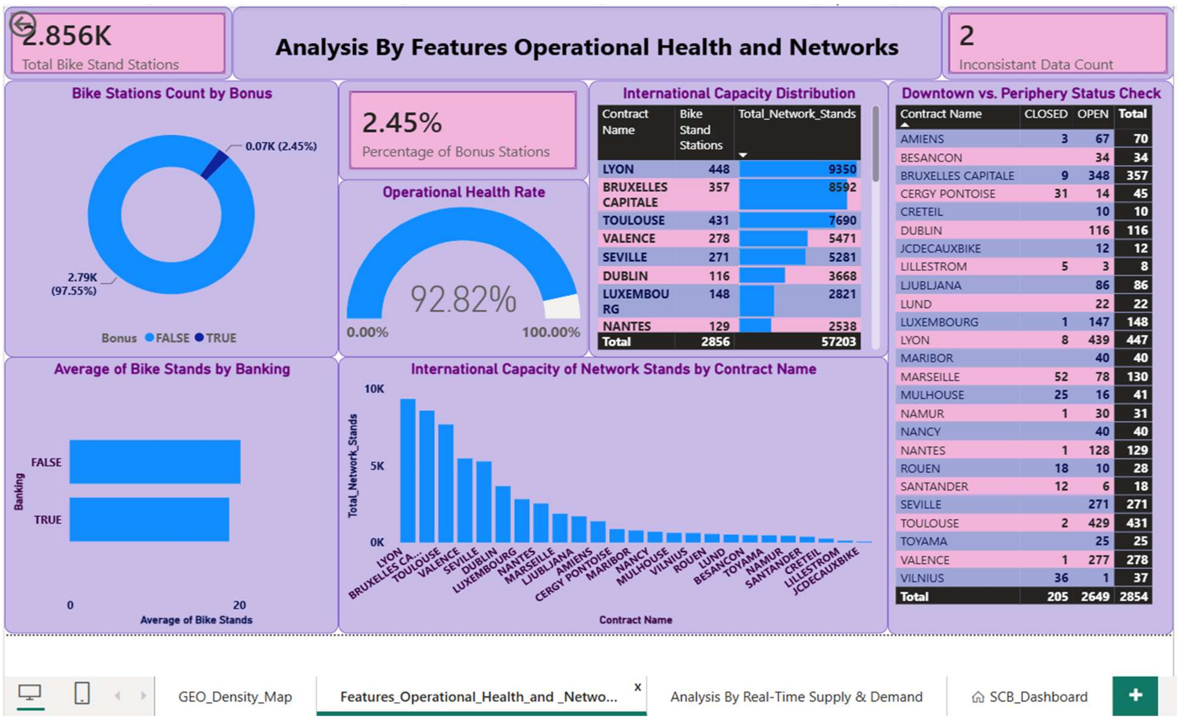
Dashboard Features:

- **Multiple Visualizations based on problem statement:**
  - Map is used to analyse visualise the number of Bike stations clustered around particular geographical location (Latitude, Longitude)
  - Column chart, Bar chart, Donut chart, Gauge, Cards, Table, and Matrix were used to analyse and visualise the data by Features, Operational health and Networks.
  - Clustered column chart, Matrix and Funnel were used to analyse and visualise the data by Real time supply and Demand.
  - Drill-down, Filters, slicers and Bookmarks were used to create interactive Reports and Dashboard.

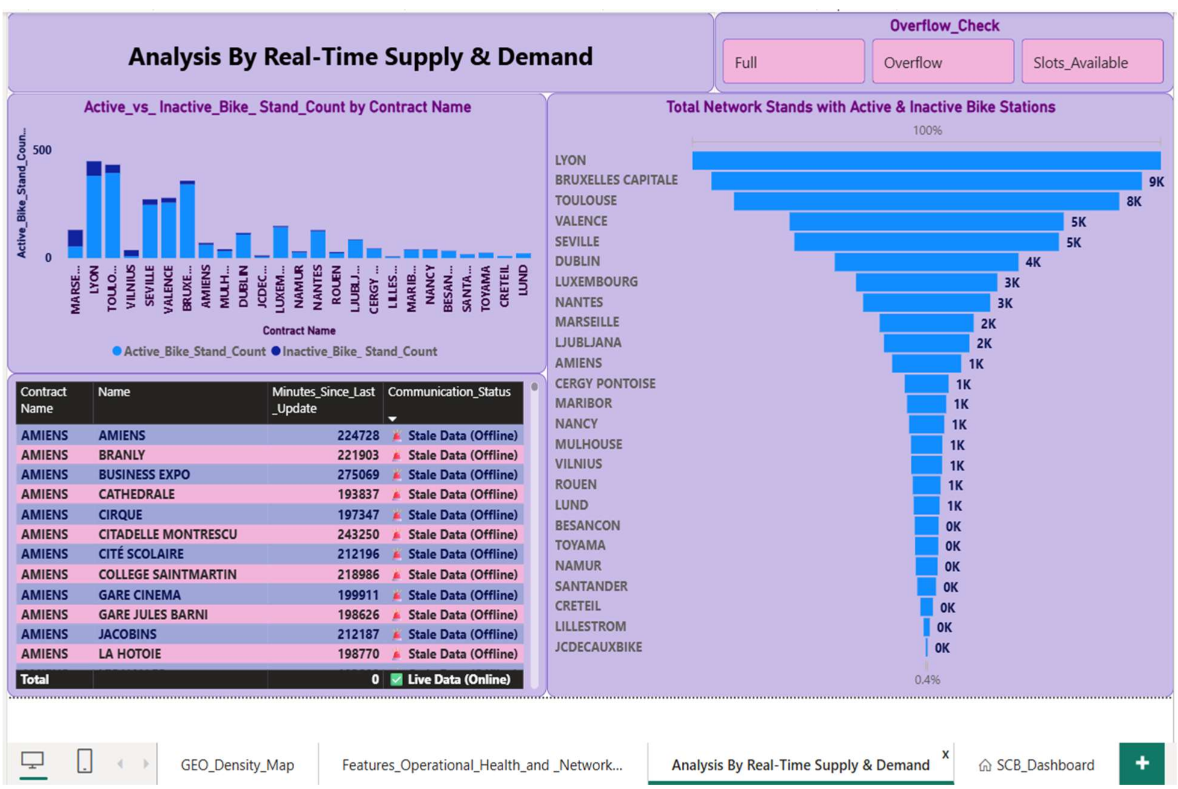
Dashboard



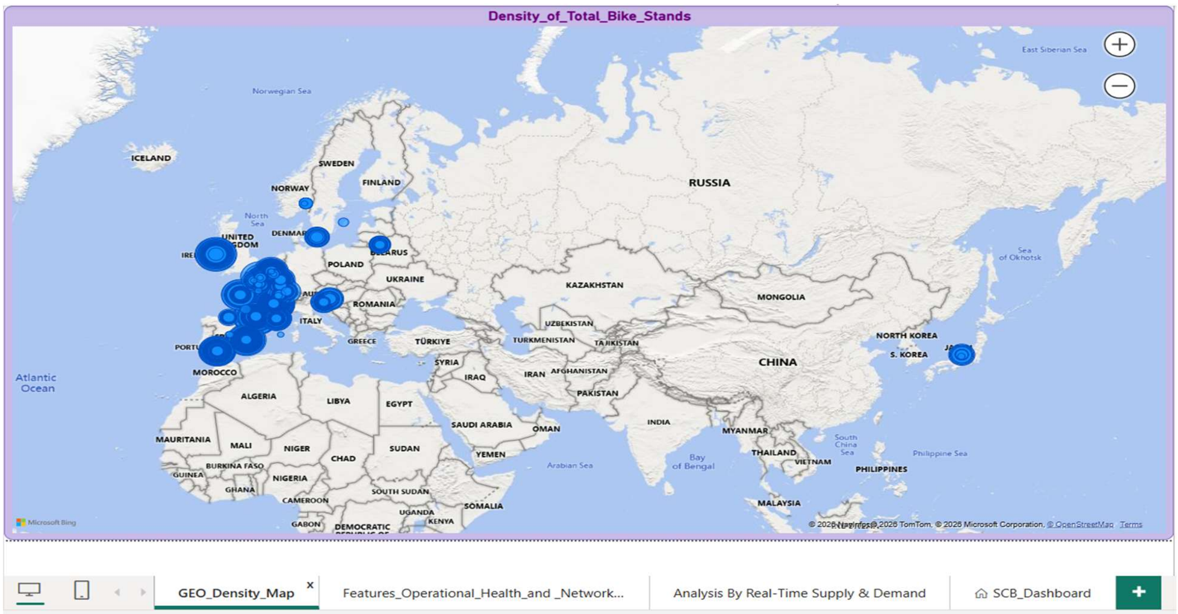
Analysis By Features, Operational Health & Networks



Analysis By Real Time Supply & Demand



GEO_Density Map



9. Insights

- ### Key Findings:

  Network Architecture & Top Clusters

  - **Total Network Stations:** ~3,000 active locations
  - **Top High-Density Cluster Capacity:** 1,236 total bike stands (Toulouse urban center)
  - *Station 1 (43.60, 1.45):* 439 stands
  - *Station 2 (43.61, 1.45):* 407 stands
  - *Station 3 (43.61, 1.44):* 390 stands

  Incentive & Terrain Management

  - **Bonus Location Distribution:** 2.45% of the entire network
  - **Total Active Bonus Stations:** 57 stations (targeted at extreme geographic choke points/inclines)

  Hardware & Asset Capacity

  - **Average Capacity (Terminal-Free Stations):** 20.19 stands per station
  - **Average Capacity (With Banking Terminals):**
  - 18.85 stands per station
  - **Capacity Increase via Digitization:** +1.34 stands on average (+7.1% capacity growth per terminal-free station)

🔧 Operational Health & Maintenance

- **Global Operational Health Rate:** 92.82% fully functional
- **Global System Outage Rate:** 7.18% offline
- **Total Offline Infrastructure:** 205 stations out of service
- **Target Operational Threshold:** >95.00% (Industry Gold Standard)
- **Critical Isolation Radius:** 1.5 km (High-priority threshold for geographically isolated offline stations)

🌍 Flagship Market Concentrations

- **Top 2 Flagship Contract Capacity:** 17,942 total bike stands
- *#1 Lyon, France:* 9,350 total stands
- *#2 Bruxelles-Capitale:* 8,592 total stands

- ## Analysis insights:

1) When evaluating individual hot spots, the data reveals that our infrastructure distribution is not driven by single, massive mega-stations. Instead, it relies on highly optimized, high-density networks of mid-sized stations grouped closely together.

Our top cluster is located at coordinates (43.60, 1.45) (the vibrant urban center of Toulouse, France), commanding 439 bike stands. It is immediately backed up by a neighboring station at (43.61, 1.45) with 407 stands and another at (43.61, 1.44) with 390 stands. Within just a few hundred meters, this single city zone unlocks over 1,200 total stands, proving that this region is a primary engine for our transit network.

2) Our infrastructure analysis reveals that exactly 2.45% of the entire station network is designated as a 'Bonus' location. Out of nearly three thousand active stations, this premium incentive is highly exclusive, operating as a scalpel rather than a sledgehammer to regulate our fleet.

A 2.45% distribution indicates a strategy of hyper-targeted asset management. It proves that our network does not suffer from widespread terrain or elevation issues across all of Europe. Instead, the system faces highly localized, extreme geographic choke points.

These 57 bonus stations are likely positioned on a few specific, severe inclines—such as the historic hill districts overlooking cities like Lyon or Toulouse. By limiting rewards strictly to these steep zones, the business avoids diluting the value of user credits while still protecting the most vulnerable high-altitude stations from running empty.

3) Our asset capacity analysis reveals a surprising trend: stations without physical banking terminals are larger on average (20.19 stands) than those with credit card hardware (18.85 stands).

App-only or membership-card stations require no heavy outdoor computing hardware. They are entirely digital. This allows the business to scale dock capacity up to 20+ stands effortlessly and cheaply wherever neighborhood demand spikes, without worrying about the added cost of a terminal. Comparatively Physical banking terminals are expensive to purchase, install, secure, and maintain.

This data proves that our network successfully drives its heavy volume through mobile application adoption. The system does not need to rely on old-school, on-site payment booths to justify building large, high-capacity stations.

Moving forward, the business can confidently expand neighborhood grid capacities using lightweight, terminal-free docks, reserving banking hardware strictly as a minor accommodation for transient tourist zones.

4) Our network is operating at a 92.82% operational health rate. While nearly 93% of our European footprint is fully functional and ready for riders, this leaves roughly 7% of our infrastructure—equivalent to 205 stations—completely out of service across our grid.

In micro-mobility, a 7% system outage doesn't just mean empty docks; it creates network blind spots. If a user rides a bike to their destination only to discover that their neighborhood station is part of the 7% offline pool, they experience immediate transit friction. This leads to abandoned bikes, dropped rides, and a direct loss in casual user confidence.

To push our operational health past the industry gold-standard threshold of 95%, we must move away from a global view and isolate the root causes:

- Seasonal/Contractual Auditing: We need to investigate if these 205 closed stations are evenly distributed anomalies or if they are concentrated in a specific city contract. A highly concentrated shutdown implies a localized system software upgrade, a targeted vandalism wave, or a planned seasonal winterization freeze.
- Prioritized Re-activation: By cross-referencing our earlier proximity analysis, we should instantly prioritize reopening any offline stations that are geographically isolated. A closed station with no neighbors within 1.5 km hurts the network far more than a closed station in the middle of a high-density cluster."

5) When evaluating our international capacity distribution, our operational footprint is anchored by two massive flagship contracts. Lyon, France, firmly commands the #1 spot as the largest infrastructure network in our portfolio with 9,350 total bike stands. It is closely tailed by Bruxelles-Capitale at 8,592 stands.

These two regions represent the absolute heavyweights of our micro-mobility network, acting as the primary revenue and volume drivers for the entire enterprise.

From a business perspective, the combined total of 17,942 stands between these two cities represents our core asset concentration. Because Lyon and Brussels hold such an overwhelming percentage of our physical docks, any shift in their performance drastically sways our global metrics. This is exactly why our earlier findings are so critical: if a portion of our 7.18% offline anomalies (the 205 down stations) are sitting within these two massive contracts, fixing them immediately in Lyon or Brussels will yield a far higher financial return than repairing stations in our smaller tail-end expansion markets

6) While running out of bikes halts trips before they begin, running out of stands breaks the back-end of the user journey. When a station registers 0 available bike stands, it creates an immediate crisis of convenience. Users are trapped with an active rental asset they cannot return, leading to immense user frustration and illegal locking on city street furniture.

- gridlocked stations heavily clustered around major central transport hubs (like the train stations in Brussels or Lyon) during morning rush hours. Hundreds of suburban commuters flow inward simultaneously, overwhelming the local dock grid.
- The Downhill Inundation: Cross-reference these stations with your Elevation or Location Zones. If these stations sit at the very bottom of a steep hill, it proves that gravity is pulling your fleet down into a natural pooling basin faster than your 2% bonus incentives can convince people to ride them back up.

This watchlist provides a highly actionable roadmap for physical infrastructure adjustments and truck routing:

- Micro-Rebalancing Dispatches: Fleet trucks should be dispatched to clear bikes out of these specific top addresses just before peak commute windows open, artificially creating a 'buffer' of empty stands.
- Dock Expansion Capital: If the same addresses top this list month after month, it indicates a structural capacity failure. We should permanently expand the physical footprint of these specific stations by adding 5 to 10 more docks, funding this expansion by scaling back capacity at the 'Ghost Stations' identified in our deficit analysis

10. Conclusions

- **Target Flagships First:** Maximize ROI by prioritizing immediate maintenance and rebalancing in the high-yield Lyon and Brussels contracts.
- **Eliminate Blind Spots:** Deploy targeted routing to reactivate the 205 offline stations, prioritizing those geographically isolated by over 1.5 km.
- **Scale Digitally:** Drive future expansion via cost-effective, terminal-free docks to capture an immediate 7.1% average capacity increase.
- **Resolve Gridlock:** Relieve zero-vacancy transit hubs through proactive truck dispatches and targeted 5-to-10 dock physical expansions.